# 4.10.12

EE25BTECH11011 - Navya

**Question**:
The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

**Solution:**
The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \tag{1}$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \tag{2}$$

Form a single matrix equation (all three lines):

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 5 & 4 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \\ 0 \end{pmatrix}. \tag{3}$$

To solve, take the first two rows and solve by inverse:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \end{pmatrix} \tag{4}$$

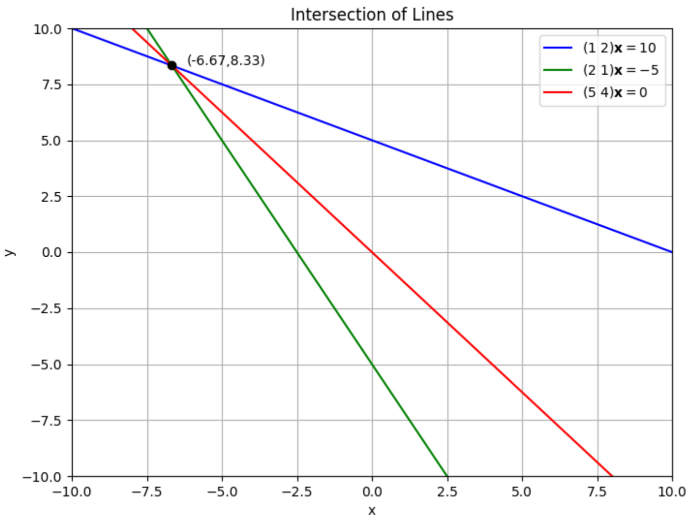
$$\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 10 \\ -5 \end{pmatrix} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \tag{5}$$

Now verify that this vector satisfies the third row of (3),

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \tag{6}$$

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} = \frac{1}{3}(-100 + 100) = 0 \tag{7}$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.


Intersection of Lines
(-6.67,8.33)
(1 2)x = 10
(2 1)x = −5
(5 4)x = 0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x
y